higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 3

11 MARCH 2019

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MARKING GUIDELINES FOR EXAMINERS

1. Examiners are reminded that the subject Carpentry and Roof Work lends itself to broad interpretations and therefore broad ranges of answers.

2. The marking guideline serves as a guideline to markers/examiners and is not intended to be absolute or prescriptive.

3. The answers provided in this memorandum are therefore not exhaustive and examiners/markers are requested to use their discretion when marking. Answers provided by students should be assessed within the context of (the students) life experiences.

4. Content should be marked and not grammar.

QUESTION 1: HIPPED ROOFS

1.1	1.1.1	Deemed to satisfy		
	1.1.2	Rationally designed		
	1.1.3	Rationally designed		
	1.1.4	Deemed to satisfy		
	1.1.5	Rationally designed		
	1.1.6	Deemed to satisfy	(6 × 1)	(6)
1.2	• Identification of trusses • Bracing details • Truss spacing		(3 x 1)	(3)
1.3	1.3.1	Adjust/Increase the truss spacing accordingly		
	1.3.2	Clean off excess mortar from the top of the walls		
	1.3.3	Adjust/Increase the eaves overhang accordingly	(3 × 2)	(6)
1.4	Drill a hole through the wall plate. Then use a masonry bit to drill a corresponding hole into the brick and fix with a suitable nail anchor.		(3 x 1)	(3)
1.5	Self-bracing occurs when the design of the roof eliminates the need for diagonal bracing.		(2 x 1)	(2)
1.6	Girder truss			(2)
1.7	• Half truss • Jack truss • Hip truss		(3 x 1)	(3)
1.8	• Wind conditions • Poor workmanship • The load on the roof • The size of the roof • The shape of the roof		(5 x 1)	(5)
				[30]

QUESTION 2: ERECTING AN L-SHAPED ROOF

2.1
- Carry trusses in an upright position.
- Do not bend the trusses.
- Stack trusses flat above the ground.
- Always cover trusses if stored outside. (4 x 1) (4)

2.2 A water-pipe level (1)

2.3
2.3.1 True
2.3.2 True
2.3.3 False
2.3.4 False
2.3.5 False
(5 × 1) (5)

2.4
- The girder truss carries the combined weight of all the valley trusses.
- The girder truss is the longest truss. (2 x 1) (2)

2.5
2.5.1 It carries/supports the jack trusses.
2.5.2 On the girder truss bottom chord
(2 × 1) (2)

2.6 Profiled sheet metal (1)

2.7 200 mm (1)

2.8
2.8.1 an internal
2.8.2 50 mm
2.8.3 tie beams
2.8.4 nails and bolts
2.8.5 grout
(5 × 1) (5)

2.9
2.9.4
2.9.9
2.9.3
2.9.1
2.9.8
2.9.7
2.9.6
2.9.5
2.9.2
(9 × 1) (9)
[30]

QUESTION 3: INSPECTION OF ROOF STRUCTURES

3.1	• Timber sizes and grades • Connector plate sizes and positions	(2 × 2)	(4)
3.2	• The correct spacing • The proper fixing method • The correct joining method	(3 x 1)	(3)
3.3	• A latent defect is seen after a period of time. • A patent defect is immediately visible.	(2 x 2)	(2)
3.4	• Timber warping over time • Roof sagging	(2 × 1)	(2)
3.5	• Timber split due to poor workmanship • Roof anchors not tight	(2 × 1)	(2)
3.6	The home builder		(1)
3.7	The engineer responsible for the design		(1)
			[15]

QUESTION 4: ROOF CLADDING

4.1	4.1.8 4.1.4 4.1.3 4.1.6 4.1.9 4.1.10 4.1.1 4.1.2 4.1.5 4.1.7	(10 × 1)	(10)
4.2	• Pigment to match the colour of the tiles • Bonding liquid improves adhesion	(2 x 1)	(2)
4.3	4.3.1 300 mm 4.3.2 450 mm 4.3.3 600 mm	(3 × 1)	(3)

4.4	• The roof pitch • The wind speed • The height of the roof • The length of the roof slope • The type of building • The terrain category		(6 x 1)	(6)
4.5	4.5.1	fascia		
	4.5.2	roof plane		
	4.5.3	purlins		
	4.5.4	in close contact		
			(4 × 1)	(4)
				[25]
			TOTAL:	**100**

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